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BACHELOR OF SCIENCE IN COMPUTER SCIENCE

PROJECT REPORT

PROJECT TITLE: CSC 489

DESIGN AND CONFIGURATION OF A SECURE HIERARCHICAL
NETWORK ARCHITECTURE WITH CISCO PACKET TRACER.

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JUNE 2026

CERTIFICATION

This is to certify that this report entitled “DESIGN AND CONFIGURATION OF A SECURED HIERARCHICAL NETWORK ARCHITECTURE WITH CISCO PACKET TRACER” is the original work of LONGHO VITALIS NGONG with the registration number SC22A828, student at the department of computer science at the University of Buea. All borrowed ideas and materials have been duly acknowledge by means of references and citations. The report was supervised in accordance with the procedure laid down by the University of Buea. It has been read and approved by:

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DECLARATION

This report entitled “DESIGN AND CONFIGURATION OF A SECURED HIERARCHICAL NETWORK ARCHITECTURE WITH CISCO PACKET TRACER” has been written by me and has not received any academic credit at this or any other institution.

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ACKNOWLEDGEMENTS

Completing this academic journey has been a profoundly enriching experience, and it would not have been possible without the unwavering support, guidance, and encouragement. I take this opportunity to express my deepest gratitude to all those who have contributed, directly or indirectly, to my growth and success. First and foremost, I will like to take this opportunity to thank God almighty for his love, mercy he showed me and the grace and strength he gave me during this academic journey. I extend my sincere appreciation to the UNIVERSITY OF BUEA for providing me with education and an environment that fosters intellectual curiosity, critical thinking, and personal development. The university's commitment to academic excellence, and diverse learning opportunities have played a pivotal role in shaping my knowledge and skills. I am truly honored to be a part of this esteemed institution. I am grateful to my lecturers, whose expertise, dedication, and mentorship have been invaluable throughout my studies. Each of them has left a lasting impact on my academic journey: DR ACHANKENG PETER L., DR DENIS L. N, DR ALI WAKA, DR NYAMSI M., DR NYANGA BERNARD, MR MAWOH R. YIRAM, MR KOMETA D., MRS TABO DELPHINE, MR MUKETE for their engaging teaching style and ability to make complex concepts understandable, for their passion for the subject and constant encouragement, for challenging me to think critically and expand my perspectives, for their patience and willingness to guide me through difficult topics. Their collective wisdom, encouragement, and constructive criticism have enhanced my academic abilities. To my incredible classmates, I owe a great deal of gratitude. The countless hours spent studying together, and supporting one another through challenges have made this journey unforgettable. Finally, and most importantly, I would like to express my deepest love and gratitude to my mother Mrs. LONGHO PRUDENCIA, and my BROTHERS LONGHO MPAFEE, YUH FABIEN AND SISTERS, LONGHO GLORY, LONGHO BESHELINE for their unconditional love, endless sacrifices, and unwavering belief in me have been my foundation throughout my academic pursuits. They have been my pillars of strength, offering guidance during moments of doubt and celebrating my achievements with immense pride. Without their encouragement, patience, and support, none of this would have been possible I would also like to acknowledge the moral support of my extended family, friends, YUH FABIEN and mentors who have encouraged me along the way. Every conversation, piece of advice, and word of motivation has contributed to my growth. As I move forward. Thank you all for being an indispensable part of my academic success

DEDICATION

I will love to dedicate this project to the University Of Buea Department Of Computer Science and to my Mother Mrs. LONGHO PRUDENCIA N.

ABBREVIATIONS

BPDU	Bridge Protocol Data Unit
CHAP	Challenge Handshake Authentication Protocol
CLI	Command Line Interface
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
IP	Internet protocol
ISP	Internet Service Provider
LAN	Local Area Network
LSA	Link State Advertisement
OSI	Open System Interconnection
OSPF	Open Shortest Path First
PAP	Password Authentication Protocol
PPP	Point-to-Point Protocol
RSTP	Rapid Spanning Tree Protocol
STP	Spanning Tree Protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
VLAN	Virtual Local Area Network
VLAN	Virtual Local Area Network
VTY	Virtual Teletype
WAN	Wide Area Network

ABSTRACT

Design and Configuration of Hierarchical Network Architecture using Cisco Packet Tracer, designing and configure of a hierarchical network architecture is based on the three-layer model: Core Layer which handles high-speed connectivity and routes traffic efficiently between networks, Distribution Layer which manages inter-VLAN routing, security policies, and redundancy, Access Layer which connects end-user devices to the network. Create a topology with appropriate devices (Layer 2 switches for Access Layer, Layer 3 switches for Distribution Layer, and routers for Core Layer). Organize the network into VLANs for segmentation. Configure VLANs on Access Layer switches. Set up trunk links between Access and Distribution Layer switches. Configure inter-VLAN routing on Distribution Layer switches. Ensure redundancy fail over using protocols like STP.

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CHAPTER ONE

INTRODUCTION

1.1 Background Study

Hierarchical network architecture is a structured design approach that organizes network devices and service to improve scalability, performance, and management. By segmenting a flat network into smaller, manageable blocks, this model restricts traffic, enhances security, and simplifies troubleshooting. In a Hierarchical model, the network is divided into distinct layers. Also known as Tiers. These layers are connected with one another in the form of a hierarchy which enables the network to be divided into more manageable blocks and these blocks limit the local traffic to remain local even if it is broadcast traffic. A hierarchical model can be applied for both LAN and WAN network design.

Hierarchical network architecture, first put forth by Cisco in 2002, has now spread throughout the industry as a recommended practice for creating dependable, scalable, and affordable networks. Early networks were flat and could only be stretched in one direction using hubs and switches, making it difficult to filter unwanted traffic and manage broadcasts. Response times would worsen as a network expanded in size. It was necessary to create a new network design, which led to the hierarchical strategy. Even while flat network designs are still used today, they are typically used for very small networks or designs that aim to reduce costs by utilizing a minimal number of routers or switches. A network is divided into layers by a hierarchical structure, and each layer has a set of functions that specify how it fits into the overall network. This gives a network designer the ability to select the best hardware, software, and

features to fulfill a specific purpose for that network layer. Additionally, data management is far more effective. Local traffic in a hierarchical design only goes to a higher tier when it is destined for another network. [1]

Hierarchical Network Design is now considered to be the best practice industry-wide to design networks that are reliable, resilient, scale-able, and also cost effective. Initially, networks were designed in a flat topology where the end devices were connected using Hubs and Switches. In order to add more devices or more users, more Switches/Hubs were added to the network. This Flat network design would cause a delay in the network if in case the network grows and also because of the use of Hubs and Switches it would be very difficult for the admins to control and limit the broadcast or filter the undesired traffic in network [2]

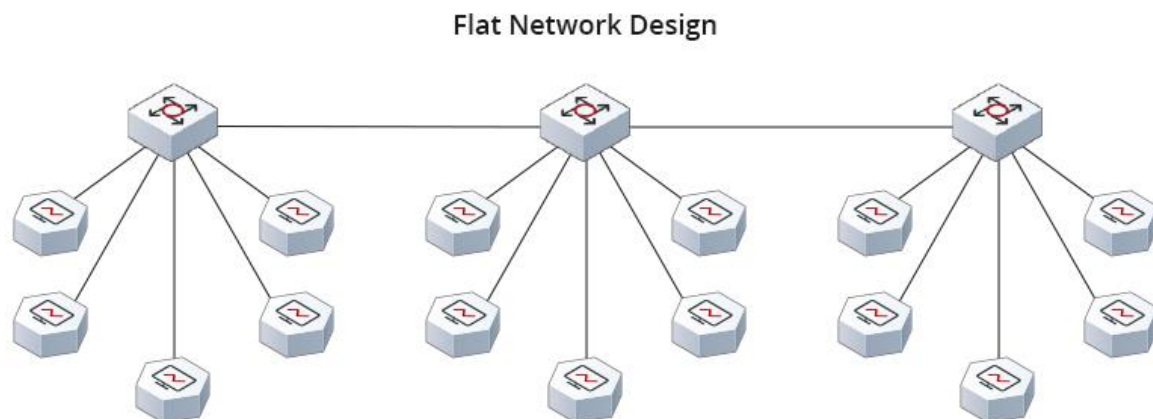


Figure 1.1: Flat Network Design [2]

Due to the above limitations of the Flat Network Design, a Hierarchical Network Design Model was introduced. In a Hierarchical model, the network is divided into

three distinct layers. These layers are connected with one another in the form of a hierarchy which enables the network to be divided into more manageable blocks and these blocks limit the local traffic to remain local even if it is broadcast traffic. A hierarchical Model can be applied for both LAN and WAN Network design.[3]

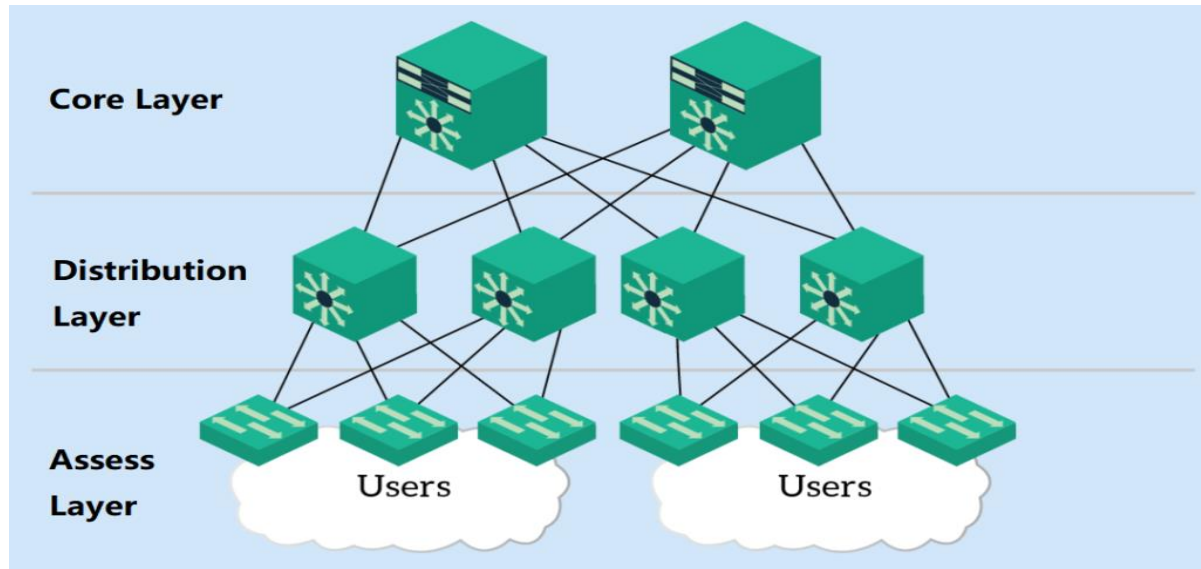


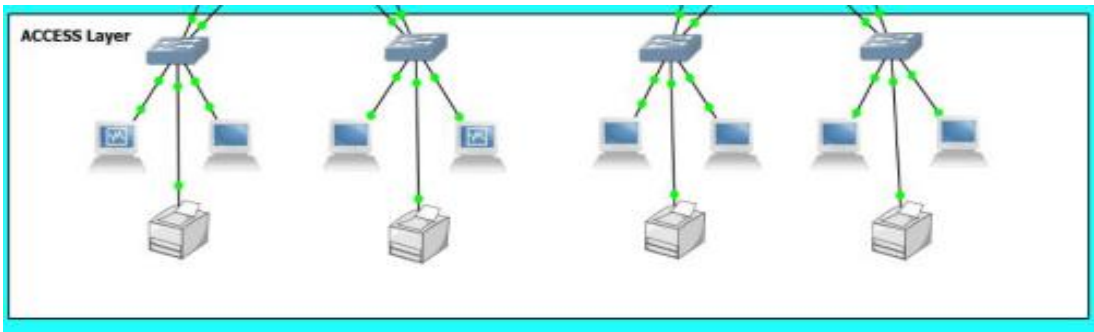
Figure 1.2: Hierarchical Network Design [4]

A 3-tier hierarchical network topology consist of three layers.

1.1.1 Access Layer:

- The Access Layer commonly referred to as the network edge is where the end-user devices connect to the network.
- It provides high-bandwidth connectivity.
- It provides Layer 2 Switching capabilities.
- Services like Port Security, Quality of Service are used in this layer

Below show the diagrammatic representation of access layer switch connectivity with end devices like the printers, pc.



-

Figure 1.3: Access layer design [1]

1.1.2 Distribution Layer:

This layer basically provides policy-based connectivity and acts as a boundary between the Access Layer and the Core Layer. Data Filtering and Routing take place in this layer.

- The Distribution Layer is mainly responsible for collecting/aggregating data from the Switches of the Access Layer and distributing it to the rest of the network.
- It acts as a border as well as a connector to both the Access Layer and the Core Layer.
- The use of routing services like OSPF.
- It provides Redundancy and Load Balancing.



Figure 1.4: Distribution layer design [1]

1.1.2 Core Layer:

This layer is often considered to be the Backbone of the network which provides fast transport between the switches present in the Distribution Layer of the network. The

Core Layer is considered to be the Backbone of the network and acts as an aggregation point for multiple networks.

- It consists of high-speed network devices responsible for switching packets as fast as possible.
- It provides interconnectivity between the Distribution Layer devices.
- It provides reliability and fault tolerance to maximize performance.

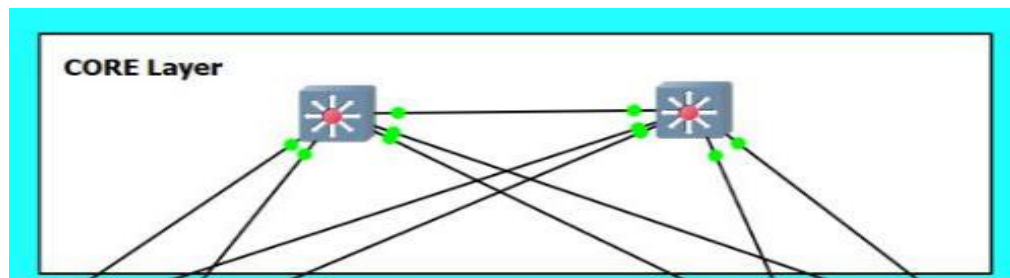


Figure 1.5: Core layer design [1]

This 3-Tier Hierarchical Network Design maximizes performance, scalability, and network availability and minimizes costs. However, Small Scale Enterprise Networks are not large enough, and therefore they use 2-Tier Hierarchical Network Design also known as. Collapsed Core Networks.

1.2 Collapsed Core Network Design:

A Collapsed Core is when the functions of the Distribution Layer and the Core Layer are implemented by a single device. This type of Network Design is used by Small Scale Enterprises to implement their networks as their network is not large enough, and they might be unable to bear the high cost of the network devices. Therefore, these Enterprises use the 2-Tier Hierarchical Network Design to reduce the network cost while maintaining most of the benefits, functions, and services offered by the 3-Tier Networks. [4]

CHAPTER TWO

LITERATURE REVIEW

2.1 Conceptual Framework

The hierarchical network design model, often referred to as the three-tier model, was popularized by Cisco Systems and has become the standard approach for designing enterprise networks. According to Tanenbaum and Wetherall (2011), hierarchical models simplify network design by breaking large, complex systems into smaller, manageable layers with well-defined roles and responsibilities. [5]

2.1.1 Three layers of hierarchical network architecture.

- **Core Layer:** This is the backbone of the network. Its primary function is to provide fast, reliable transportation of data between distribution layer devices. The core layer is optimized for high-speed switching and routing. Redundancy is critical at this layer, as failure here affects all parts of the network. Devices at the core layer are typically high-performance routers or multilayer switches.
- **Distribution Layer:** This layer serves as an aggregation point for the access layer and enforces policies such as routing, filtering, and Quality of service. It connects the access layer to the core and handles inter VLAN routing, and traffic management. Multilayer switches or routers are commonly deployed here.
- **Access Layer:** This is the point at which end devices such as PCs, printers, and IP

Phones connect to the network. Access layer switches enforce security policies, provide VLAN segmentation, and support Power over Ethernet. Port security and STP are commonly configured at this layer.

The benefits of the hierarchical model include; Modularity (each layer can be upgraded independently), scalability (the network can grow without redesigning the whole architecture), and manageability (each layer has a distinct function, making troubleshooting more straightforward).

2.1.2 Open Shortest Path First (OSPF)

OSPF is a link-state routing protocol defined in RFC 2328 by the Internet Engineering Task Force (IETF). Unlike distance-vector protocols such as RIP, OSPF maintains a complete topological map of the network in a database called the Link-State Database (LSDB). Each router exchanges Link State Advertisements with its neighbors to populate this database, and then independently runs Dijkstra's SPF algorithm to compute the shortest path to each destination.

OSPF supports a two-level hierarchical structure of its own: the backbone area (Area 0) and non-backbone areas. All non-backbone areas must connect to Area 0, either directly or through virtual links. This design reduces LSA flooding and limits the size of routing tables within each area. Area Border Routers (ABR) connect multiple areas and summarize routing information between them. [6]

2.1.2.1 Key OSPF concepts include:

- Router ID: A unique 32-bit identifier for each OSPF router, usually the highest IP address on a loop-back interface.
- Hello Protocol: OSPF routers discover neighbors by exchanging Hello packets. Adjacencies are formed when parameters such as Hello/Dead intervals, Area ID, and authentication match.
- Designated Router (DR) and Backup Designated Router (BDR): On multi-access networks (e.g., Ethernet), OSPF elects a DR and BDR to reduce the number of LSA exchanges. All routers form adjacency only with the DR and BDR.
- OSPF Cost: The metric used by OSPF is called cost, calculated as 100 Mbps divided by the interface bandwidth. Lower cost paths are preferred.
- OSPF States: Routers transition through several states when forming adjacency: Down, Attempt, Init, 2-Way, ExStart, Exchange, Loading, and Full. [7]

2.1.3 Point-To-Point Protocol

PPP is a Layer 2 protocol standardized in RFC 1661. It is designed to encapsulate network layer packets over serial point-to-point links, providing a standardized method for two directly connected devices to communicate. PPP replaced the older SLIP (Serial Line Internet Protocol) due to its richer feature set, which includes:

- Multi-protocol support: PPP can carry multiple network layer protocols such as IP, IPX, and Apple Talk simultaneously.
- Error detection: PPP uses a 16-bit CRC for error checking.

- Authentication: PPP supports PAP (Password Authentication Protocol) and CHAP (Challenge Handshake Authentication Protocol). PAP sends credentials in plaintext, while CHAP uses MD5 hashing for a more secure exchange.
- Link Control Protocol (LCP): Used to establish, configure, and test the data link connection.
- Network Control Protocol (NCP): Used to establish and configure different network layer protocols.

CHAP authentication works through a three-step process:

1. The authenticator sends a challenge message
2. The peer responds with a hash value computed using the challenge and a shared secret, and
3. The authenticator compares the response with its own computation. If they match, access is granted.[8]

2.1.4 Spanning Tree Protocol (STP)

STP was developed by Radia Perlman in 1985 and later standardized as IEEE 802.1D.

Its fundamental purpose is to prevent Layer 2 loops in networks with redundant paths.

Without STP, a broadcast frame entering a loop would be forwarded indefinitely, saturating all network bandwidth and causing a broadcast storm. [9]

STP operates by electing a Root Bridge, which becomes the logical center of the spanning tree. All other switches compute a shortest path to the root bridge and block any redundant paths. The STP process involves:

- Root Bridge Election: The switch with the lowest Bridge Identifier (BID), composed of a priority value (default 32768) and the switch's MAC address, is elected as the root bridge.
- Root Port Selection: Each non-root switch selects a Root Port, which is the port with the lowest cost path to the root bridge.
- Designated Port Selection: On each network segment, the switch with the lowest path cost to the root bridge has its port designated as the Designated Port.
- Port States: STP ports transition through Blocking, Listening, Learning, and Forwarding states. Convergence to the Forwarding state can take up to 50 seconds in standard STP (IEEE 802.1D).

Rapid Spanning Tree Protocol (RSTP), defined in IEEE 802.1w, improves convergence time to less than one second by introducing new port roles and eliminating some of the listening and learning states.

2.2 Theoretical Framework

2.2.1 OSI Reference MODEL

The Open Systems Interconnection (OSI) model, developed by the International Organization for Standardization (ISO), provides a conceptual framework for understanding how different network protocols interact. It consists of seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. The protocols discussed in this project operate at different OSI layers. Understanding the OSI model helps engineers troubleshoot by isolating problems to a specific layer, ensuring the correct protocol is examined.

Protocol	OSI Layer	Function
STP	Layer 2 (Data Link)	Loop prevention in switched networks
PPP	Layer 2 (Data Link)	WAN encapsulation and authentication
OSPF	Layer 3 (Network)	Dynamic routing and path selection
IP	Layer 3 (Network)	Logical addressing and packet forwarding
Ethernet	Layer 2 (Data Link)	Frame transmission on LAN links

2.2.2 Network convergence Theory

Network convergence refers to the time it takes for all routers or switches in a network to agree on a consistent view of the topology after a change (such as a link failure). Fast convergence is critical for minimizing service disruption. OSPF achieves relatively fast convergence compared to distance-vector protocols because it floods link state changes throughout the area immediately upon detection. STP convergence in its original 802.1D form is slow (30-50 seconds), which motivated the development of RSTP and other improvements.

2.2.3 Redundancy and Fault Tolerance

Redundancy in network design refers to the use of multiple paths, devices, or links so that a single failure does not bring down the entire network. However, redundancy at Layer 2 (switches) without STP creates loops. Redundancy at Layer 3 (routers) is handled by OSPF ability to route around failed links. The interplay between STP and OSPF in a hierarchical design ensures both Layer 2 and Layer 3 redundancy operate correctly without interference.

2.3 Empirical Literature

2.3.1 Related studies

Oluwadare and Babatunde (2019) conducted a study on OSPF performance in a simulated enterprise network using Cisco Packet Tracer. Their findings showed that OSPF converged faster than RIP in networks with more than 10 routers and provided

more accurate routing in the presence of multiple paths. They recommended OSPF for any network with more than two routers.

Musa et al. (2020) compared different WAN encapsulation protocols including PPP and HDLC in a simulated environment. Their results showed that PPP with CHAP authentication provided a significantly more secure connection than HDLC or PPP without authentication, with negligible performance overhead. They concluded that CHAP should be the default configuration for all PPP serial links in security-sensitive environments. [10]

Eze and Nwachukwu (2021) investigated the impact of STP on network performance in a topology with multiple switches and redundant paths. Their simulation demonstrated that enabling STP eliminated all broadcast loops within 45 seconds of network startup, while RSTP reduced this to under 2 seconds. The study recommended RSTP over legacy STP for modern enterprise deployments.

Abubakar et al. (2022) designed a hierarchical campus network for a Nigerian university using Cisco Packet Tracer, integrating OSPF and VLANs. Their study highlighted that the hierarchical model improved traffic management and allowed network administrators to isolate faults more efficiently than a flat design. They also noted that proper IP addressing and OSPF area planning were critical success factors.

These empirical works collectively support the design choices made in this project: using OSPF for dynamic routing, PPP with CHAP for secure WAN links, and STP for

loop-free switching. They also validate the use of Cisco Packet Tracer as a reliable simulation tool for undergraduate network projects.

2.3.2 Gap in the literature

Despite the extensive literature on individual protocols, few studies have documented the simultaneous integration of OSPF, PPP, and STP within a single hierarchical network model with step-by-step configuration commands and verification outputs. Most existing works focus on one protocol at a time or use flat topology. This project fills that gap by providing an end-to-end, multi-protocol hierarchical network implementation, complete with configuration steps and verification evidence. [11]

CHAPTER THREE

METHODOLOGY AND DESIGN

3.1 Methodology

Designing a hierarchical network requires a systematic approach to ensure scalability, reliability, and efficiency. These methodology include planning, designing, implementing, and testing networking systems. And as a result the phases are:

1. Analysis Phase: Requirements were gathered by identifying the network's functional goals, including the number of users, device types, routing requirements, WAN connectivity, and redundancy needs. The three-tier hierarchical model was selected as the architectural framework.
2. Design Phase: Based on the requirements, a logical network topology was created. IP addressing schemes were planned for all interfaces, sub-nets were allocated for each layer, and protocol parameters were determined for OSPF, PPP, and STP.
3. Simulation Phase: The physical and logical topology was constructed in Cisco Packet Tracer. All devices were placed, connected with appropriate cable types, and assigned IP addresses.
4. Configuration Phase: Each device was configured via its CLI (Command Line Interface). OSPF was configured on all routers, PPP with CHAP was configured on serial links, and STP was configured on all switches.

5. Verification Phase: The network was tested using Cisco IOS show commands and connectivity tests (ping and trace route) to verify that the protocols were functioning correctly.
6. Documentation Phase: All configuration steps, commands, and outputs were documented with screenshots and explanations.

This iterative approach ensures that each phase builds on the previous one, and problems identified during verification are fed back into the design or configuration phases for correction.

3.2 Tools and Software

TOOL/SOFTWARE	VERSION	PURPOSE
Cisco packet tracer	9.0.0.0910	Primary networking simulation environment
Microsoft word/WPS	2013	Project documentation
Cisco IOS	9.0.0.0810	Router switches operating system
Windows 10 Pro OS	10.0.0.19045	Host operating system for simulation

3.2.1 Justification for Cisco Packet Tracer

Cisco Packet Tracer was selected as the primary simulation tool for the following reasons:

- It is freely available to students through the Cisco Networking Academy.
- It supports all required Cisco IOS commands for OSPF, PPP, and STP configuration.

- It provides a visual, drag-and-drop interface for topology design.
- It allows real-time simulation and observation of packet flows.
- It is widely recognized in academic and professional circles as a valid network design tool.

3.3 Network Specifications

3.3.1 Functional requirement

DEVICE	MODEL	QUANTITY	LAYER	ROLE
router	Cisco 2911	3	core	OSPF routing, PPP,WAN links
Multilayer Switch	Cisco 3560	2	Distribution	Inter-VLAN routing, OSPF
Access Switch	Cisco2960	3	access	STP,VLAN access
PC end device	Generic pc	10	access	End user Simulation
Server	Server PT	2	access	DNS/DHCP Simulation
Printer	Printer PT	2	Access	Documentation print out
Access point	Access point-PT	1	access	Wireless connection
Connection type	Copper straight-through	16	Access	Connect dissimilar device
	Copper cross-over	7	Access	Connect similar device
	Serial DCE	2	core	Relay or carrier of data

3.4 DESIGN

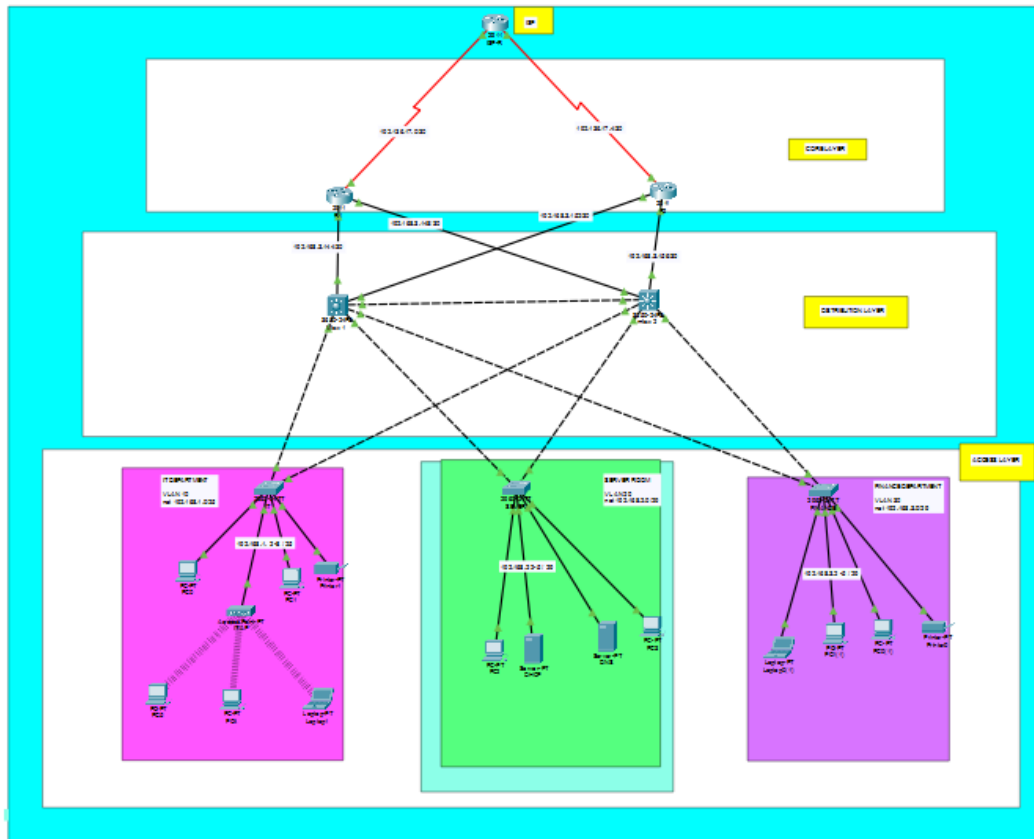


Figure 3.1 show Design of a hierarchical network architecture.

3.4.1 The Three-Tier Structure

3.4.1.1 CORE LAYER TWO ROUTERS (R1&R2)

At the core layer the router (R1) and router (R2) connect to each other through an OSPF, Point to Point link forming the backbone. They route to the internet .Each router connect to both multilayer switches through a primary link and a redundant cross link so that if one router or cable fails, traffic will still flow.

Serial Data Communication Equipment cable is used to connect two router each linking to Internet service provider and each line having the network IP address of 192.136.17./30 (192.136.17.19 & 192.136.17.19) and the other having the network 192.136.17.4/30. (192.136.17.4.6 & 192.136.17.4.5)

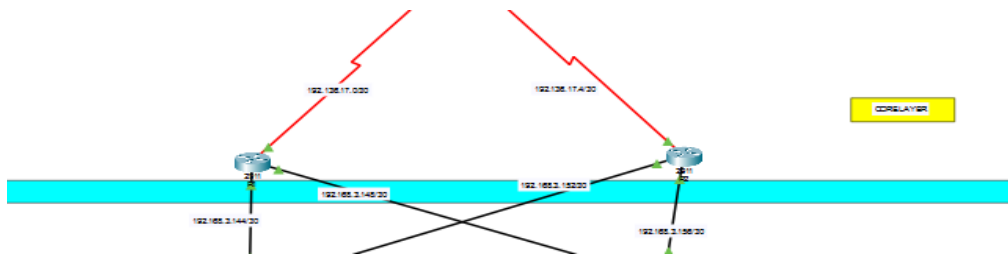


Figure 3.2 show core layer two routers (r1&r2)

3.4.1.2 DISTRIBUTION LAYER (MLS1 & MLS2)

This is the heart of the design. The two multilayer switches links to each other by trucking and carrying all the VLANs which link to the router with the network IP address of 192.168.3.156/30(192.168.3.158 & 192.168.3.157) ,192.168.3.152/30 (192.168.3.153 & 192.168.3.154) and of 192.168.3.144/30(192.168.3.146 & 192.168.3.145) , 192.168.148/30 (192.168.149 & 192.168.150) respectively. The link up to the router using four link and link down to the access layer with six links all with copper cross-over cable. The multilayer switch one (MLS1) is the STP Root Bridge and active gateway. MLS2 is the standby for failover with no manual intervention.

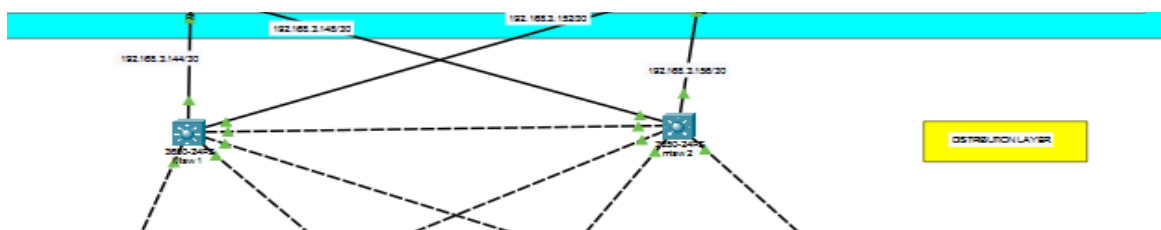


Figure 3.3 show distribution layer (mls1 & mls2)

3.4.1.3 ACCESS LAYER (IT DEPARTMENT, SERVER ROOM, FINANCE DPT)

Each access switch is connect to active MLS1 and MLS2 with two link each but STP blocks the redundancy port to prevent loops if the primary link dies, the block port activates with second.

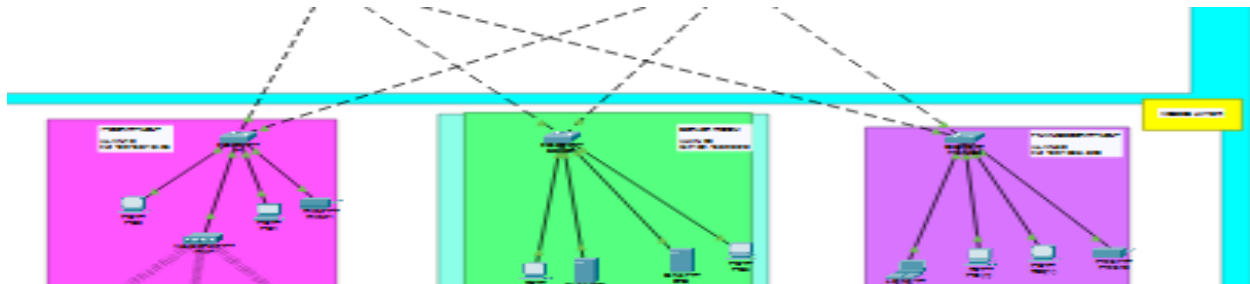


Figure 3.4 show access layer (IT Department, Server Room, Finance department)

3.5 REASON FOR THIS DESIGN

- No single point of failure where every devices has a redundant path.
- OSPF handle automatic re-routing if a link breaks
- HSRP ensure end devices never lose their default gateway
- STP prevents broadcasting storms from all the redundant loops.

In the layer, the following are the IP address and VLANs with their subnet mask are:

IP addresses for VLAN 10 are 192.168.1. 3/25, 192.168.1 4/25 192.168.1. 5/25, 192.168.1. 7/25 192.168.1. 8/25

IP addresses for VLAN 20 are 192.168.2. 2/29, 192.168.2. 3/29, 192.168.2. 4/29 192.168.2. 5/29.

IP addresses for VLAN 30 are 192.168.3. 2/29, 192.168.3. 3/25, 192.168.3. 4/25 192.168.3. 5/29.

3.4 Project Timeline (Gantt chart)

The following table represents the project timeline over a 127-days period:

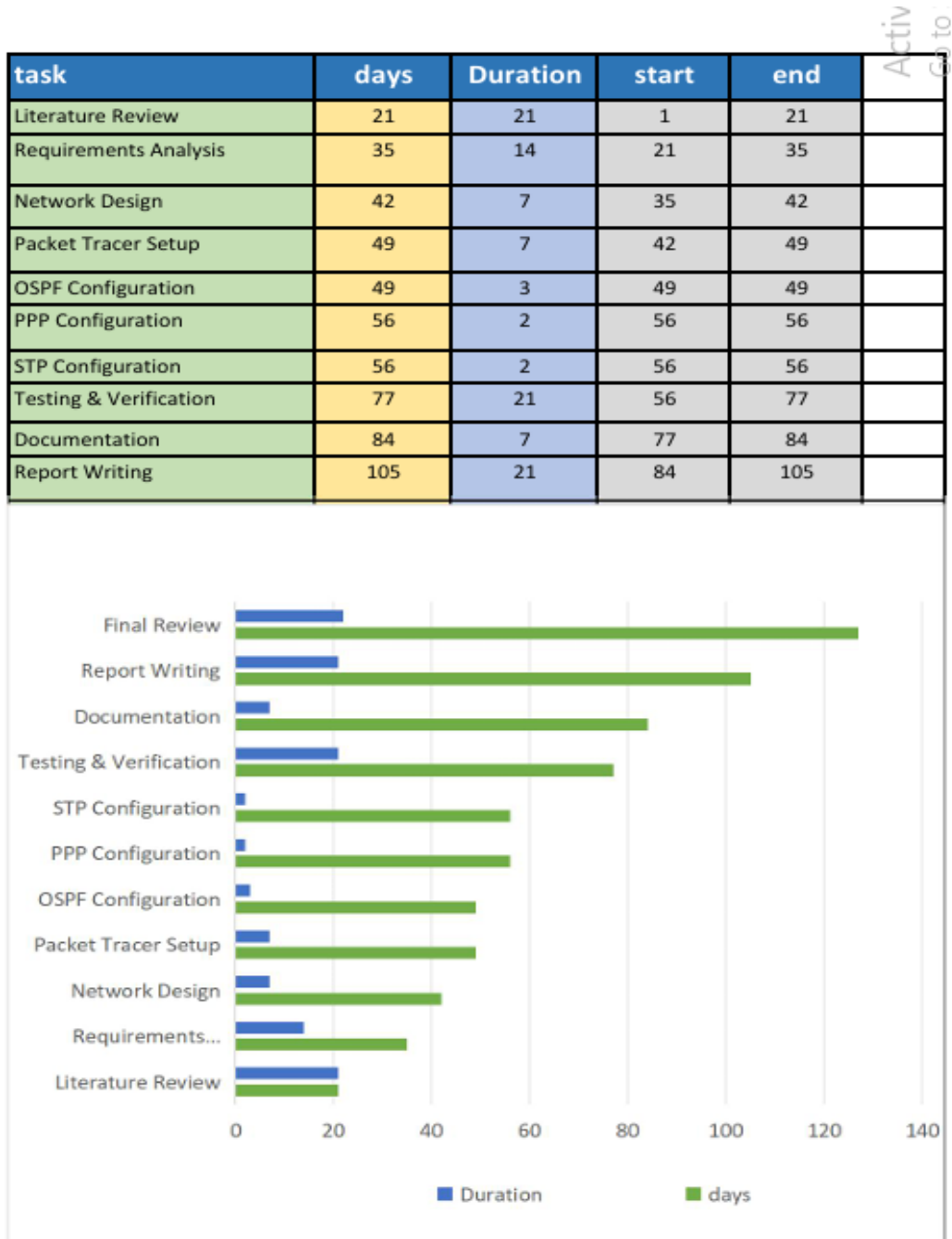


Figure 3.5 show the project timeline over 127-days period

CHAPTER FOUR

IMPLEMENTATION

This chapter presents the step-by-step implementation of the hierarchical network, including device configurations, special commands, and descriptive placeholders for screenshots taken during the simulation in Cisco Packet Tracer.

4.1 Network Topology Design

The network topology was constructed in Cisco Packet Tracer with three core routers (R1, R2, R3) connected via serial WAN links, two distribution multilayer switches (DMSw1, DMSw2), and three access switches (ASw1, ASw2, ASw3,). End devices (PCs printers, servers & access layers) are connected to access switches. The topology mirrors a small enterprise campus network.

4.1.1 Cable Types Used:

- Serial DCE/DTE cables between R1-R2 and R2-R3 for WAN links (PPP encapsulation).
- Straight-through Ethernet cables between routers/switches and end devices.
- Cross-over cables between switches, multi switch, router.
- Gigabit Ethernet uplinks between distribution and core layers.

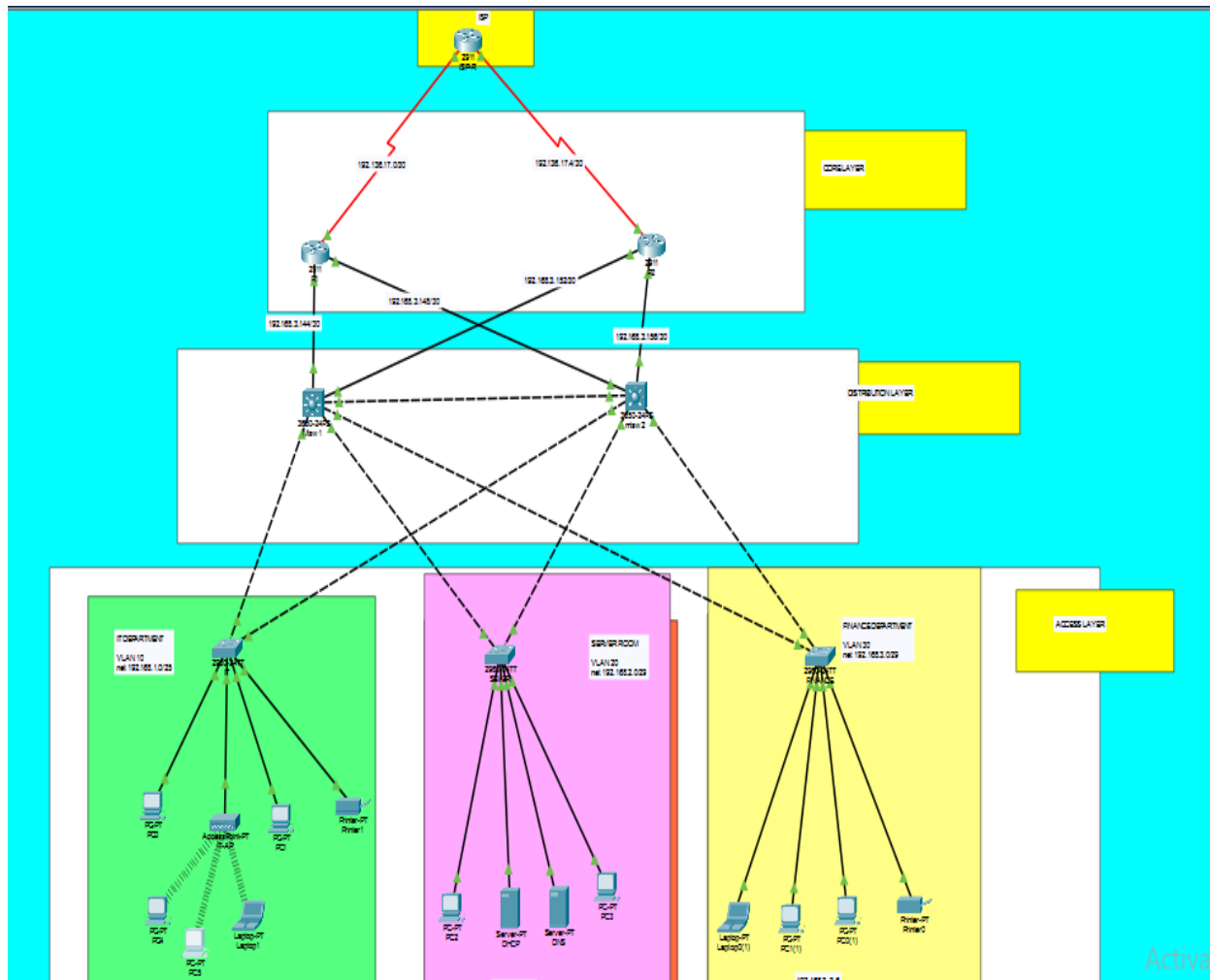


Figure 4.1 *Network Topology Design of a hierarchical network*

4.2 BASIC DEVICE CONFIGURATION

4.2.1 Hostname and Password Configuration (All Devices)

Each device was configured with a hostname, encrypted enable password, and message-of-the-day (MOTD) banner. Below is the screenshot showing the configuration for RouterR1

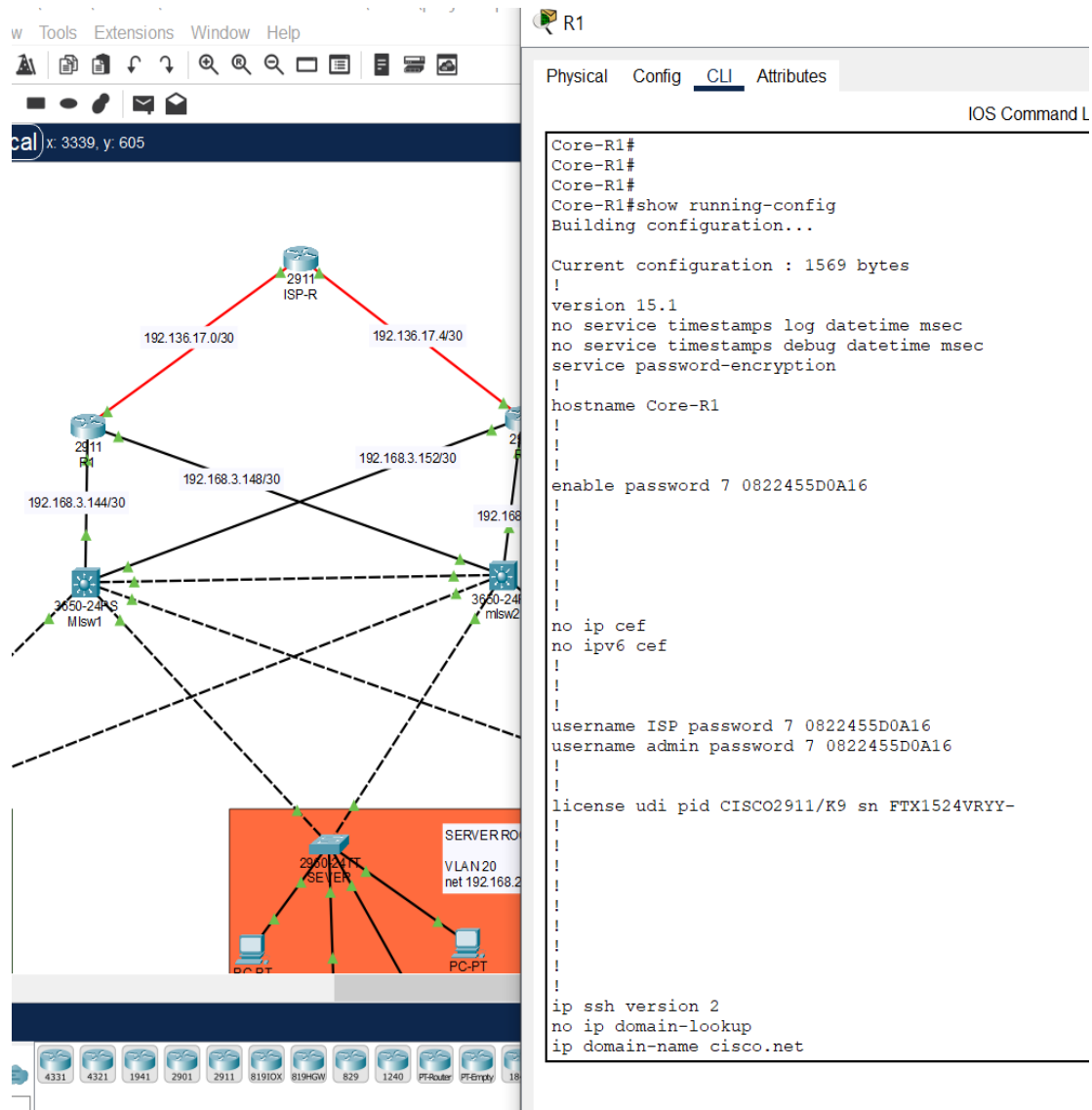


Figure 4.2: CLI of R1 showing hostname and password configuration commands entered successfully

4.2.2 Interface IP Address Configuration (Router 1)

Router R1 has three interfaces: GigabitEthernet0/0 (connected to DS1), Serial0/2/0 (connected to R2) for OSPF Router ID.

2\saves\project1.pkt

R1

Physical Config CLI Attributes

IOS Command Line Interface

```

ip ssh version 2
no ip domain-lookup
ip domain-name cisco.net
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
ip address 192.168.3.146 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 192.168.3.150 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
!
interface Serial0/2/0
ip address 192.136.17.1 255.255.255.252
encapsulation ppp
ppp authentication chap
no keepalive
clock rate 64000
!
interface Serial0/2/1
no ip address
clock rate 2000000
!
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/1
no ip address
clock rate 2000000
!

```

192.136.17.4/30

192.168.3.152/30

192.168.3.150

3650-24T msw2

SERVER0

VLAN20

net 192.168.2

PC-PT

29 1240 PT-Router PT-Empty 18

Figure 4.3: show IP interface on R1 showing all interfaces with correct IP addresses assigned

4.2.2 Interface IP Address Configuration (Router 1)

Router R1 has three interfaces: GigabitEthernet0/0 (connected to MSW1), Serial0/2/0 (connected to R2), and Loopup0 (for OSPF Router ID and PPP).

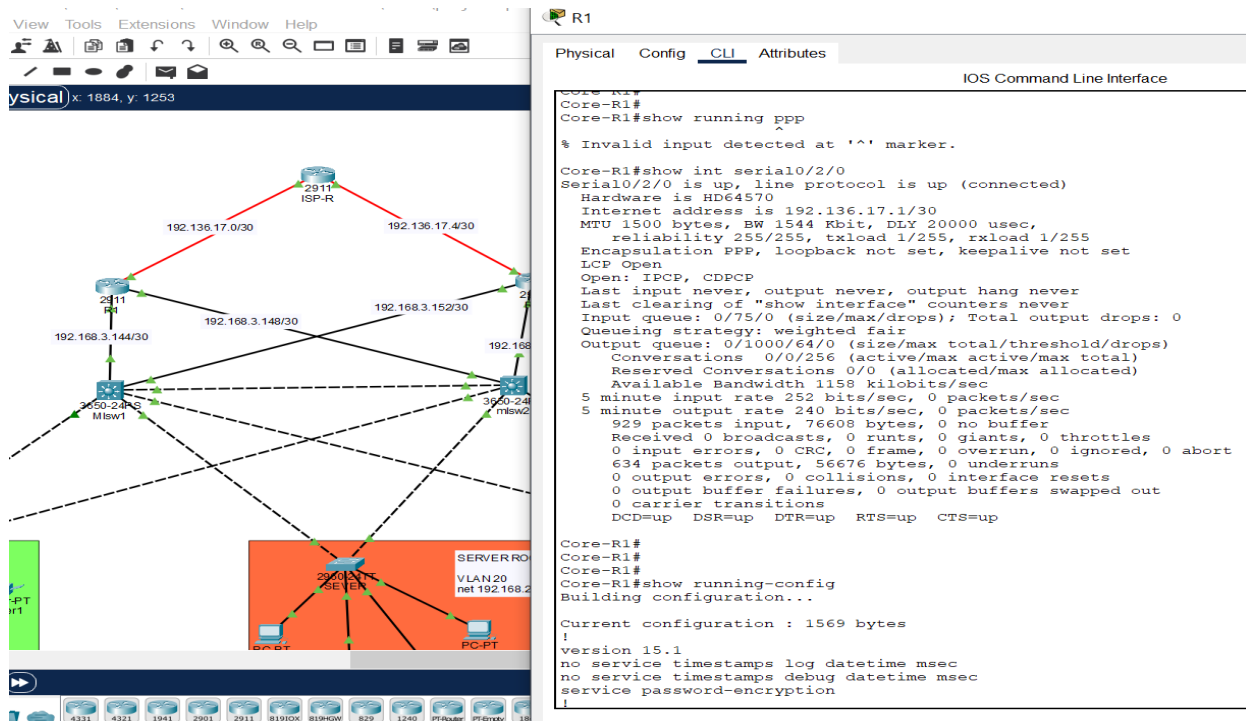
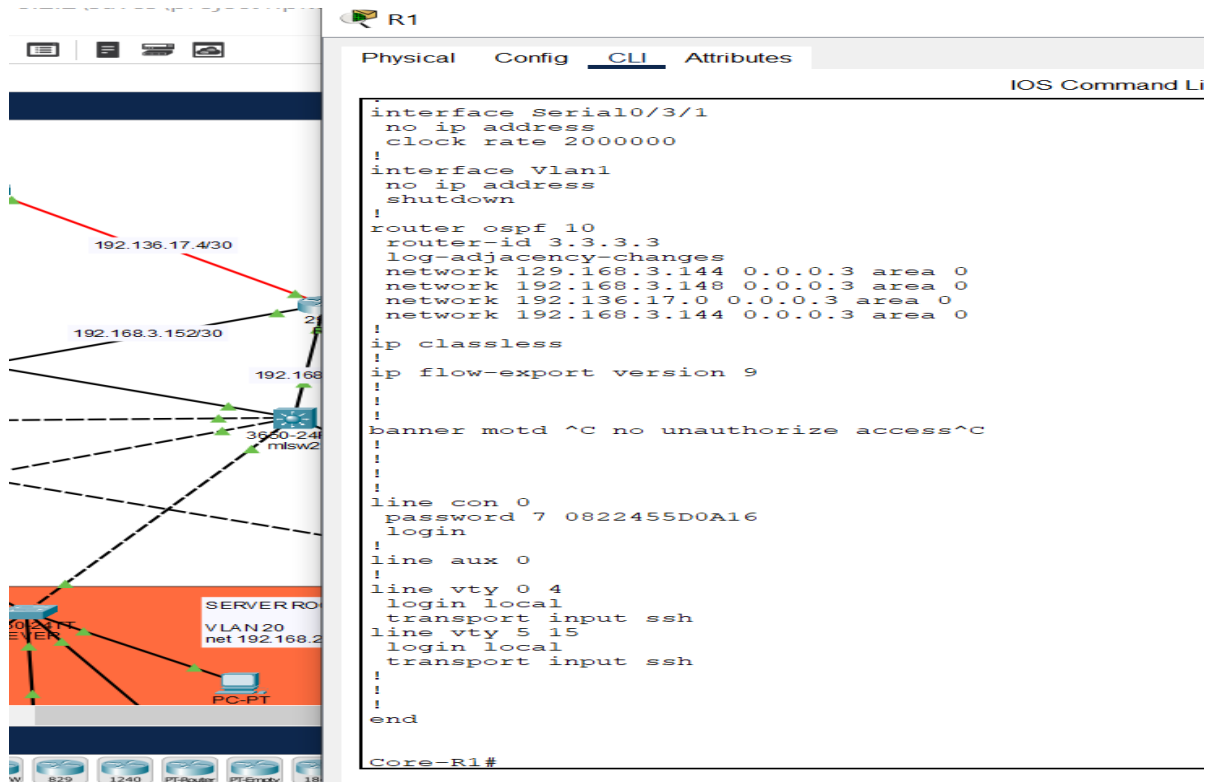


Figure 4.4: *show IP interface brief on R1 showing all interfaces with correct IP addresses assigned]*

4.2.3 VLAN Configuration on Access switches (ASw4)

VLANs are created on the distribution and access switches to segment user traffic.

ASw4 carries VLAN 30

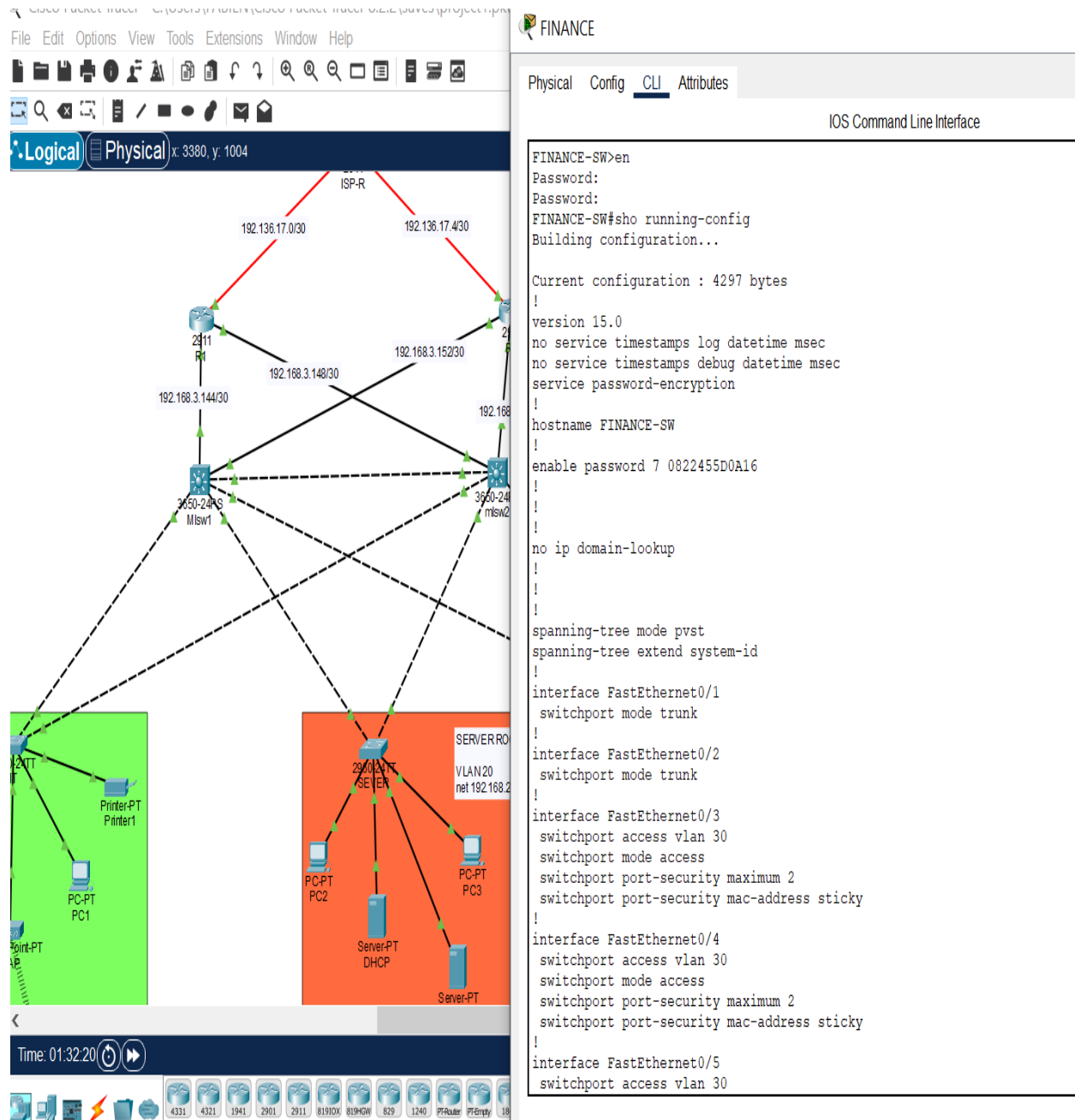


Figure 4.5: *show VLAN brief* on ASw4 showing VLAN 30 created and assigned to ports Fa0/1–Fa0/24

PPP is configured on the serial WAN links between routers. CHAP authentication is used for security. Both sides of the link must have matching usernames and passwords configured for CHAP to succeed.

4.3.3 TCP Configuration

STP is already enabled by default on Cisco switches. However, manual configuration of the Root Bridge and port priorities is required to ensure a deterministic and optimal spanning tree topology. DS1 is configured as the primary root bridge, and DS2 as the secondary root bridge.

4.3.4 Fast Ethernet port on Access Switch Port

Fast Ethernet port is configured on access layer switch ports connected to end devices to bypass the listening and learning states, allowing PCs to connect immediately.

4.4: TESTING

Ping tests were conducted from PC1 (VLAN 10, IT Department) to PC0 (1) (VLAN 30, FINANCE Department) and to pc2 (VLAN 20, SEVER ROOM) to verify end-to-end reachability across the entire hierarchical network:

4.5 VERIFICATION

Traceroute was run from PC0 to PC5 to confirm the path taken through the hierarchical network:

```
C:\>tracert 192.168.3.6

Tracing route to 192.168.3.6 over a maximum of 30 hops:

  1  0 ms      1 ms      0 ms      192.168.1.1
  2  *         12 ms     704 ms     192.168.3.6

Trace complete.

C:\>tracert 192.168.3.5

Tracing route to 192.168.3.5 over a maximum of 30 hops:

  1  0 ms      0 ms      34 ms     192.168.1.1
  2  *         12 ms     0 ms      192.168.3.5

Trace complete.

C:\>
```

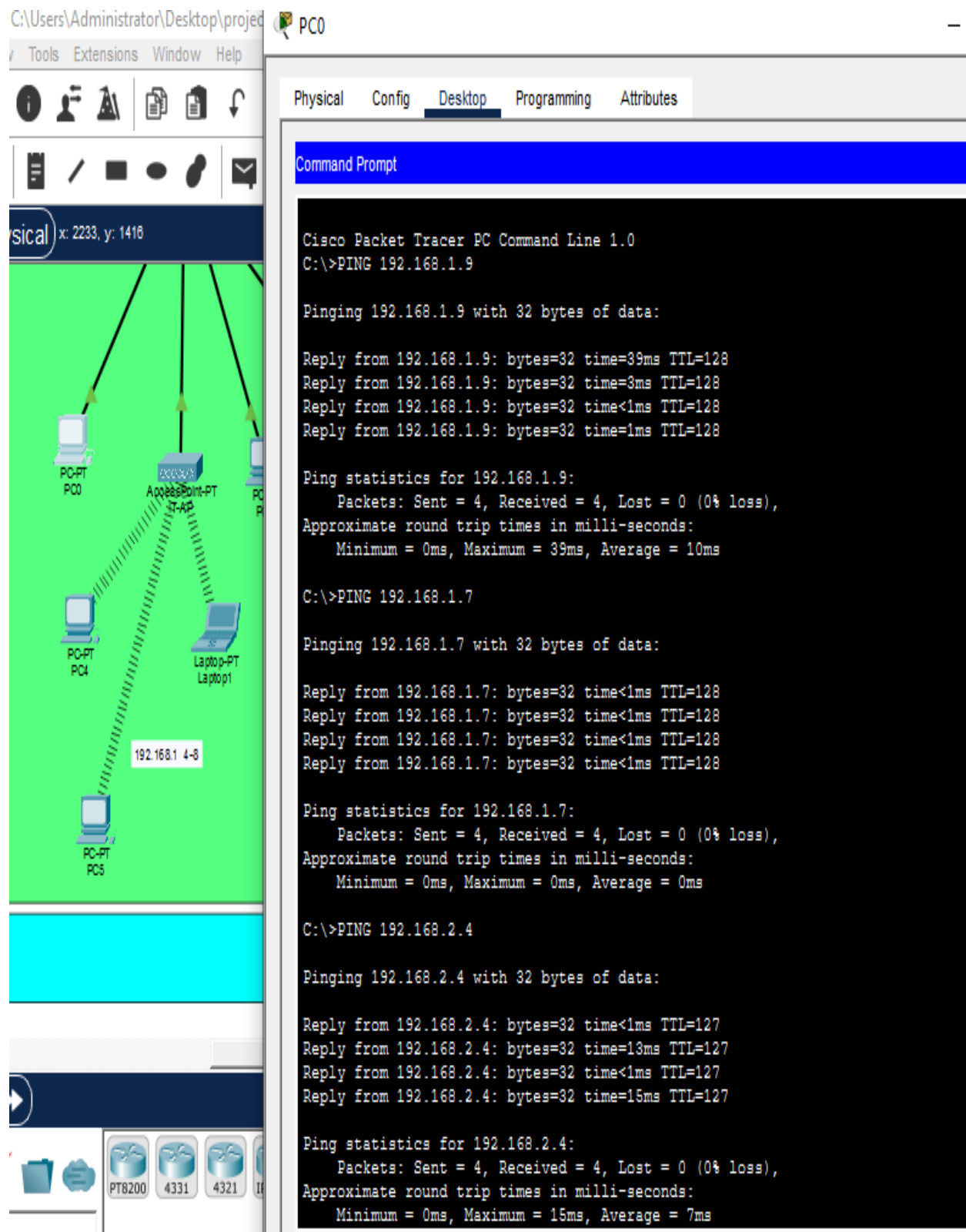


Figure 4.5: *show testing from VLAN 10, VLAN 20, VLAN 30 end devices*

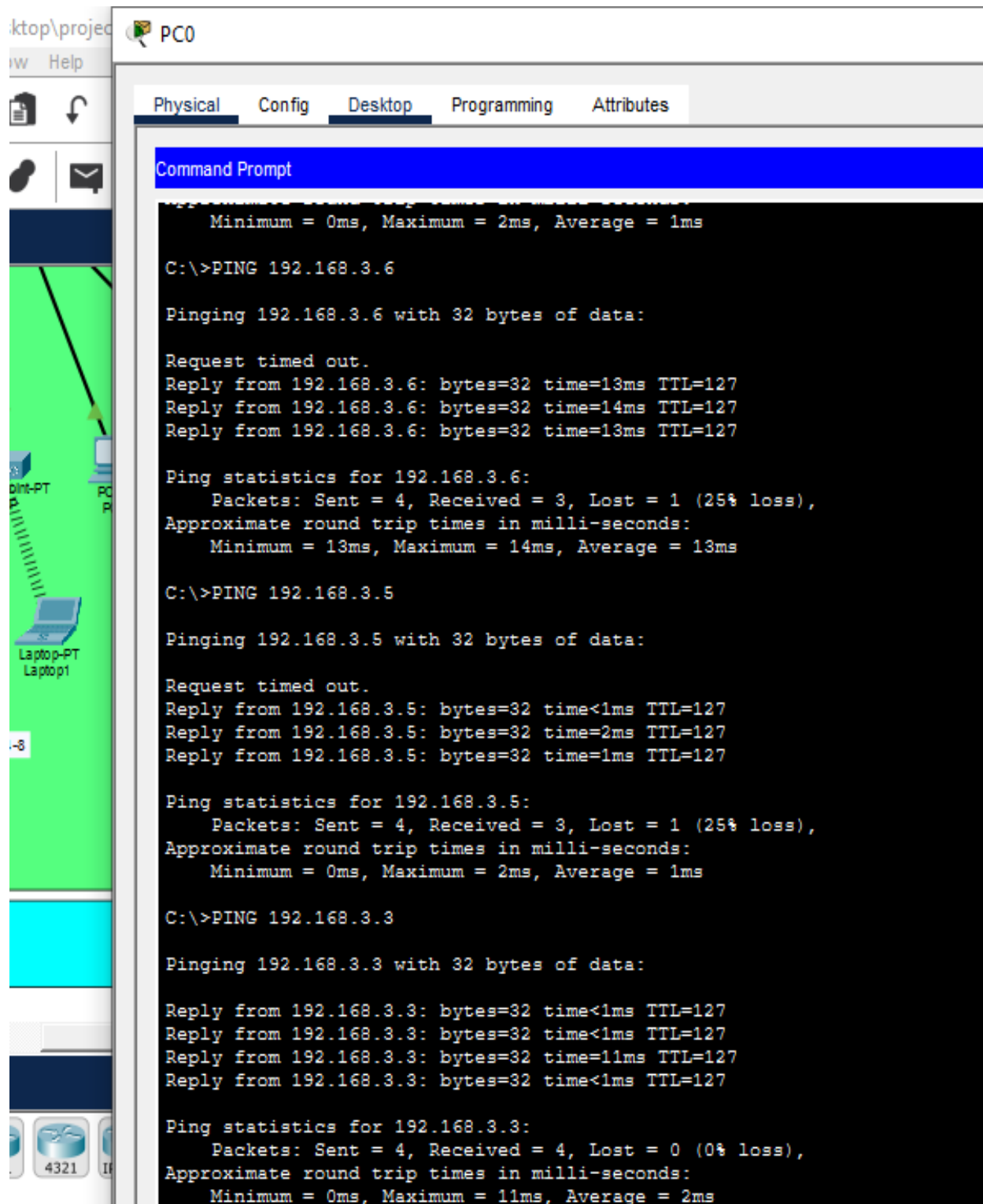


Figure 4.5: show testing from IT department on pc0 and all other end devices on server room, financial department.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSION

This project has successfully demonstrated the design, simulation, and configuration of a hierarchical network architecture integrating Open Shortest Path First (OSPF), Point-to-Point Protocol (PPP), and Spanning Tree Protocol (STP) using Cisco Packet Tracer.

The three-tier hierarchical model, comprising the Core, Distribution, and Access layers, provided a structured, scalable, and manageable framework for network design. The separation of responsibilities across layers ensured that routing, aggregation, and end-user connectivity were handled by appropriately equipped devices, mirroring real-world enterprise network design principles.

- OSPF was configured in a multi-area setup (Area 0 and Area 1), enabling dynamic route discovery and exchange across the entire network.
- PPP with CHAP authentication was successfully implemented on all serial WAN links. The three-way CHAP handshake provided a secure, authenticated connection between routers, preventing unauthorized access through WAN interfaces. This addresses a common vulnerability in enterprise networks that use insecure or unauthenticated WAN encapsulation.

- STP was configured with deliberate Root Bridge election, ensuring a deterministic loop-free topology while maintaining physical redundancy and protecting the STP domain from unauthorized switch connections.

End-to-end connectivity was verified through ping and trace route tests, confirming that data can traverse from any end device in the access layer through the distribution layer, across the core layer via PPP WAN links, and to any other destination in the network. All protocol verifications passed, demonstrating a fully operational, multi-protocol hierarchical enterprise network.

This project fulfills all stated objectives and contributes a practical, documented reference for multi-protocol hierarchical network design in academic and professional contexts.

5.2 RECOMMENDATIONS

Based on the findings and limitations of this project, the following recommendations are made for future work and operational deployment.

Implement OSPF Authentication: To further secure OSPF adjacencies, MD5 authentication should be configured on all OSPF interfaces. This prevents rogue routers from injecting false routing information into the OSPF domain. Future research and development effort should focus on developing automated configuration tools and optimization algorithms to address the complexity challenges associated with hierarchical network.

5.3 REFERENCES

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